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2019, ICRTAC 2019Performance Evaluation of Area-Based Segmentation Technique on  
Ambient Sensor Data for Smart Home Assisted LivingR. Kavitha<sup>a\*</sup>, Sumitra Binu<sup>b</sup><sup>a</sup>*CHRIST(Deemed to be University), Bangalore, Karnataka, India*<sup>b</sup>*Independent Researcher, Bangalore, Karnataka, India*

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**Abstract**

Activity recognition(AR) is a popular subject of research in the recent past. Recognition of activities performed by human beings, enables the addressing of challenges posed by many real-world applications such as health monitoring, providing security etc. Segmentation plays a vital role in AR. This paper evaluates the efficiency of Area-Based Segmentation using different performance measures. Area-Based segmentation was proposed in our earlier research work. The evaluation of the Area-Based segmentation technique is conducted on four real world datasets viz. Aruba17, Shib010, HH102, and HH113 comprising of data pertaining to an individual, living in the test bed home. Machine learning classifiers, SVM-R, SVM-P, NB and KNN are adopted to validate the performance of Area-Based segmentation. Amongst the four chosen classification algorithms SVM-R exhibits better in all the four datasets. Area-Based segmentation recognise the four test bed activities with accuracies of 0.74, 0.98, 0.66, and 0.99 respectively. The results reveal that Area based segmentation can efficiently segment sensor data stream which aids in accurate recognition of smart home activities.

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**Keywords:** Activity Recognition; Segmentation; Smart Home; Machine Learning; Elder Care

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## 1. Introduction

Owing to the migration of younger generation to other countries in search of greener pasture, elderly people, people with disabilities and patients with dementia live independently in a socially isolated environment. The fact that these people have less control over their body movements, makes them more prone to accidents in their regular routine life. When they are living alone in a normal house, it is not easy to keep track of such accidents and take timely action. Smart home helps to monitor the people who lives alone by tracking their daily activities in a timely manner and by updating the family members with the details of activities. It also helps to inform family members or hospitals whenever there is an abnormal or emergency situation.

Smart home activity recognition system helps to identify regular human activities in a smart home ambience. In late 90's ,Oscar [1] proposed the primary work on human activity recognition. Over the past decade, there has been a substantial research on AR. Still there are many challenges related to accurate recognition of activities which motivates researchers to propose novel mechanisms for improving the accuracy of activity recognition in a practical situation. These challenges include (1) Selection of measured attributes (2) Designing of data acquisition system (3) Data collection in a practical scenario (4) Appropriate pre-processing like noise removal, sensor data representation, segmentation etc., (5) Feature selection (6) Support new user without retraining the system [6]. Area-Based segmentation aids in addressing the challenges 2, 3, and 4. Based on the sensing technology the research on activity recognition systems can be categorised in to three. The foremost one is video based sensing technology. In this sensing technology video cameras are installed in different areas of home to capture the movement of a resident. This technology requires massive storage and computation capability to process video data. It also has an impact on the privacy of the resident as surveillance cameras are installed in different areas of home to capture his/her movement. The second technology is wearable sensing technology in which the sensing mechanism is connected with the resident in the form of a wearable device such as smart watch. Sensors can also be attached to his cloth, can be worn around his/her neck embedded in a pendant, can be implanted in his body etc. The acceptability rate of this technology is very less, as patients, disabled and elders find it uncomfortable and inconvenient to put on. The final sensing technology is ambient technology in which the ambient sensors namely door sensor, motion sensor, light sensor, temperature sensor etc., are installed in different areas of the home to monitor the resident. Ambient sensors sense the ambient when the residents carry out the activity within the realms of possible range of sensors. When the residents carry out their usual work, the sensors which are installed in the home area generates the sensor events in a sequence. The performed activity is inferred from the captured sensor sequence. For example when the resident performs the "Meal\_Preparation" activity at kitchen, the ambient sensors (motion, temperature, etc.) deployed in the kitchen triggers the events. The activity recognition system will recognise this "Meal\_Preparation" activity with the help of different pre-processing techniques (like segmentation) and machine learning classifiers. Segmentation techniques has a vital role in accurately identifying the activities which are recognised by an AR system. Significant research work has happened on segmentation techniques for smart home sensor data stream to recognise an activity precisely in a practical situation. In the domain of AR, many researchers have proposed segmentation techniques such as Activity-Based segmentation [2], Time-Based segmentation [4], and Area-Based segmentation [5].

The main objective of this work is (1) to understand the taxonomy of smart home AR system and (2) validating the performance of Area-Based segmentation by applying the segmentation approach on four different datasets using various performance measures. The Area-Based segmentation was proposed by [5] in their research on activity recognition in a smart home environment. As a continuation, this research work compares the performance of Area-Based segmentation when applied to four different datasets. The rest of this paper is organised as follows: Section II reviews the taxonomy of smart home activity recognition system, section 3 discusses the methodology and algorithms used, section 4 includes discussion of results obtained on Area-Based segmentation on four different datasets and section 5 concludes the work done.

### Nomenclature

|   |                                                                  |
|---|------------------------------------------------------------------|
| A | radius of                                                        |
| B | position of                                                      |
| C | further nomenclature continues down the page inside the text box |

## 2. Taxonomy of Smart Home Activity Recognition System : Literature Review

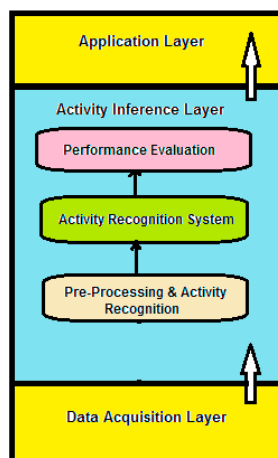


Fig. 1. Taxonomy of Smart Home Activity Recognition.

Activity recognition plays an important role in a smart home for various applications. The taxonomy of AR in a smart home can be explained in terms of the three layer architecture as depicted in Fig 1. The bottom most layer of the architecture is the Data acquisition layer. This layer handles with the designing of the smart home layout using sensor network. The ambient sensors such as temperature sensor, motion sensor, etc., are installed in the smart home to sense and measure the attributes of interest such as temperature, motion etc. The sensed data along with a sensor ID are send to the related application server for further use like visualization, analysis and monitoring of smart home resident in real time. Data acquisition layer need to address issues related to the design like choice of sensors, data collection protocol, energy utilization, and flexibility as it is dealing with the real physical units of the smart home[1]. The second layer of the architecture is the Activity inference layer. The sensed data and its corresponding IDs stored in the application server are used for activity recognition. Activity inference layer consists of three sub layers viz.Pre-Processing & Activity recognition, Activity Recognition System and Performance Evaluation. The functionalities of these sub layers can be explained as follows:

**2.1 Pre-Processing & Activity recognition :** The datasets Aruba17, HH102, HH113 and ShiB010, are used in this research work. The first three datasets which are available in the public domain are pre-processed and annotated. The dataset ShiB010, which is collected in real time environment needed pre-processing since it is ambiguous form. As a part of pre-processing, PHP code is written to decode this ambiguous form to understandable sensor events. The pre-processed data are segmented for improving accuracy of classification. Normally the segmentation happens in two ways: (a) Online segmentation where the segmentation takes place during data collection (b) Offline segmentation where the segmentation takes place after completion of entire data collection. Researchers have proposed many segmentation techniques which includes(a) Activity-Based segmentation[2] which in sensor data stream is divided into segments depending on the type of activity, (b) Sensor-Based segmentation [3] which divides equal number of sensor events from the stream of sensor data, (c) Time-Based segmentation [4] which divides the sensor data stream based on equal time duration, and (d) Area-Based segmentation[5] which divides the data sequence into segments when the triggering sequence of the sensors change. Fig 2 illustrates the first three segmentation techniques. Activity-Based segmentation and Sensor-Based segmentation are used as a segmentation technique in offline mode which happens after completion of data collection. Time-Based and Area-Based segmentation are used as a segmentation technique in online mode which happens during data collection.

**2.2 Activity Recognition System:** AR system is designed based on two approaches namely knowledge driven approach and data driven approach. In knowledge driven approach, temporal logic and description logic rules are derived to recognise the activities[6]. In data driven approach, supervised , semi-supervised and un-supervised algorithms are used to predict the activities in a smart home. Activity recognition systems are designed (i) for single Vs multi resident (ii) to find normal Vs abnormal activities, and (iii) to recognize activities in online vs offline mode.

As shown in the figure, Activity-Based segmentation divides the sensor data stream in to segments S1, S2,..... based on happened activityA1, A2,..... Sensor-Data based segmentation divides the sensor data stream in to segments S1, S2,..... by 'n' equal sensor events. Time-Based segmentation divides sensor data stream in to segments S1, S2,..... by 'n' equal seconds duration. The last two segmentation approaches are suitable for on-line activity recognition since the segmentation is not directly related to the activity. The last segmentation technique is proposed based on home layout and sensor installed location. When a person moves from one location to another, the sensors placed in the first location stops triggering and sensors in the current location starts triggering. Using this information, Area-Based segmentation divides the triggered sensor events into segments as per the change in sequence of sensors. After segmentation, suitable features are acquired from each segment. The acquired features are utilised for training and testing phases of the classification algorithm. Normally there are three types of features extracted from the sensor segments viz., frequency based, time based, and environment based. The extracted features are used to train machine learning classifiers which inturn helps to recognise the activity correctly in the test phase.

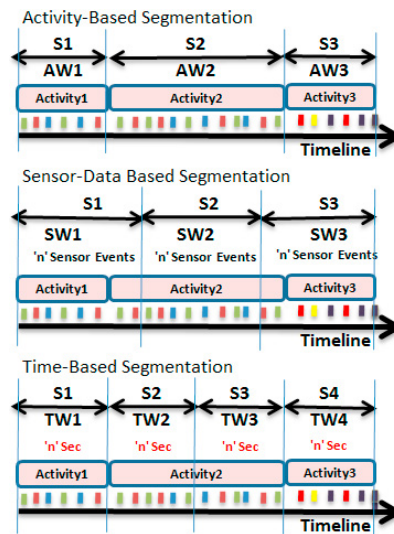


Fig. 2. Segmentation Techniques used in Ambient Sensor Data

**2.3 Performance Evaluation :** The recognised activities need to be evaluated to understand the success rate of an activity recognition approach with respect to a specific application. In case of supervised algorithms, the performance measures such as Accuracy, precision, F-Measure, ROC etc, are calculated using confusion matrix. In case of un-supervised algorithms, the performance measures including sum of square error(SSE), Recall etc., are calculated using contingency table.

The third and final layer of the architecture is an application layer which deals with real time application of a smart home. Activity recognition system is designed for different purposes like monitoring the behaviour of resident, healthcare, eldercare, to provide security and entertainment, reduce energy consumption etc.

In this work four datasets collected from real smart home test beds are used. Data are segmented using Area based segmentation which is suitable for online mode of activity recognition. Following segmentation, domain based features are extracted from each segment. Four well known machine learning classifiers are used to classify the activities and performance are validated using different accuracy measures.

### 3. Methodology

In a sensor deployed smart home, most of the daily activities of the resident are strongly connected with home area like Bed-Room, Kitchen, Living room, etc. Awareness of this association helped to propose and develop an Area-Based segmentation for smart home activity recognition. The sensors deployed in a smart home sense the environment and generate the sensor sequence as a stream of sensor triggered events in non-decreasing order of triggered time. Each sensor event consists of Date and Time at which the sensor is triggered, Sensor-Id and State (ON/OFF) of the sensor. Further, this sensor events are divided into segments.

Area-Based segmentation method is proposed based on the relationship between sensor deployed area and activity happened. For example, the areas such as Bedroom, Bathroom, etc., are strongly and directly associated with the activities "Sleeping", "Bed\_to\_Toilet", etc, because the activity Sleeping mostly happens in the Bedroom and so on. Since each home area is deployed with a distinct set of sensors, the sensor triggered data stream is divided into segments based on the particular home area where the sensor is triggered by an activity. Fig 3 shows the ambient sensor data stream and Area-Based segmentation approach. This figure illustrates the process of Area-Based segmentation which can be explained as follows. In a smart home, when the resident performs regular activities viz. Meal\_Preparation(A1) and Eating(A2) in the kitchen area(Area-1), the sensor deployed in (Area-1) start triggering and generates the sensor sequence. After completing the activities A1 and A2, resident moves to next activity A3 at Area-2. Since the resident is moving from Area-1 to Area-2, the sensor deployed in Area-1 stops triggering and the sensors deployed in Area-2 starts triggering. Later, the recorded sensor sequence is divided into segments at the change point where the triggering sequence of the sensors change. Area-Based segmentation is implemented using the concept of Pruned Exact Linear Time (PELT) [7] change point detection method.

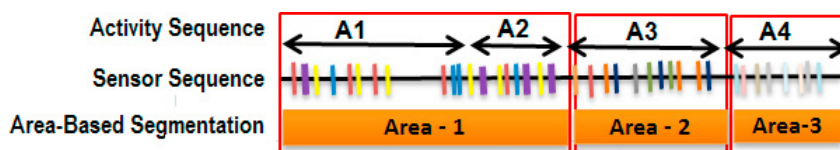


Fig 3. Area-Based Segmentation Approach on Ambient Sensor Data Stream

#### 4. Results and Discussion

In this research work, in order to analyse the performance of the Area-Based segmentation, four datasets viz. Aruba17, ShiB010, HH102, and HH113 are considered. These datasets are by-products of CASAS project [8], by Washington State University(WSU), USA. The events triggered by the sensors installed in a smart home is recorded as entries in dataset. The events generated by the normal motion sensor, wide-area sensor, temperature sensor, Door sensor, Light sensor, Light switch, and Battery level sensor are labelled with the ID "M", "MA", "T", "D", "LS", "L", and "BA" respectively. Details of the sensors deployed in all four datasets are given in the Table 1. Aruba17 test bed is installed with 40 sensors comprising of "31-Motion sensors, 5-Temperature sensors, and 4-Door sensors" and altogether these sensors have recorded 1048574 sensor events for 3656 activity occurrences. ShiB010 test bed is installed with 23 sensors comprising of "19-Motion sensors, 3-Temperature sensors and 1-Door sensor" which have jointly recorded 81647 sensor events for 327 activity occurrences. HH102 test bed is installed with 52 sensors which comprises of "23-Motion sensors, 2-Temperature sensors, 26-Light sensors and 1-Door sensor" which have jointly recorded 413142 sensor events for 2595 activity occurrences. HH113 test bed is installed with 67 sensors comprising of "23-Motion sensors, 5-Temperature sensors, 33-Light sensors and 6-Door sensors" and altogether these sensors have recorded 1046914 sensor events for 820 activity occurrences.

Each dataset is annotated with different activities. The intention of this research work is to analyse the performance of Area-Based segmentation, a set of seven activities which are common to all four dataset is considered. The name of the first activity is "Bed\_to\_Toilet" which means resident using restroom in a smart home. The other activities are "Work", "Eating", "Leave\_Home", "Hygiene", "Meal\_Preparation", "Relax", and "Sleeping" which means working in office room, eating preparing food, leaving from home, personal hygiene in bathroom, watching TV or reading book, and resting on the bed in bed room respectively.

Table 1. Details of Sensors Deployed in All Four Datasets.

| Dataset | Motion-Sensors | Temperature Sensors | Door Sensors | Light Sensors | Light Switches |
|---------|----------------|---------------------|--------------|---------------|----------------|
| Aruba17 | M001-M031      | T001-T005           | D001-D004    | -             | -              |
| ShiB010 | M001-M019      | T001-T003           | D001         | -             | -              |
| HH102   | M001-M023      | T102,T104           | D002         | LS001-LS022,  | L001-L003      |
| HH113   | M001-M023      | T101-T105           | D001- D006   | LS001-LS023   | L001-L010      |

In the domain of smart home AR, researchers generally use three types of features viz. Time domain features, Frequency domain features and Environment based features to recognize the activities. Area-Based segmentation technique is solely related to smart home environment. So, in this research work thirty environment based features like Unique sensor ID for each activity, duration of each activity, count of each unique sensor triggered during activity etc, are extracted from each segment. After feature extraction, the extracted features are used to train the machine learning classifier to build the classification model. In order to analyse the accuracy of activity recognition, different supervised machine learning algorithms are tested in this research work. Out of all, the most familiar and commonly known machine learning classifiers namely Support Vector Model using Radial basis function kernel (SVM-R), using polynomial kernel (SVM-P), Naive Bayes classifier (NBC), and k-Nearest Neighbour (kNN) exhibits commendable classification accuracy. So, in this research work SVM-R, SVM-P, NBC, and kNN are used to analyze and compare the performance of segmentation technique. To increase the utilization of available data for training and testing, 5-fold cross validation method is used. This method divides the entire dataset into 5-equal parts, consider one part at a time for testing and rest for training the model. This process is repeated for all 5 parts and the result of sub phases are combined to produce a single result. The implementation of Area-Based segmentation technique for smart home AR is done with the support of R-programming, an open source data analytics tool developed at Bell Laboratories [9]. The analysis of four machine learning algorithms indicates that SVM-R performs better w.r.t classification accuracy in all four data sets.

A confusion matrix is derived from all the four datasets to measure the classifier performance on AR using Area-Based segmentation. A confusion matrix is a table, which is used to express the performance of classifier. Using this table, the other evaluation metrics TP-True Positive, FP-False Positive, Pr-Precision, Re-Recall and F\_M-F\_Measures are derived. Normally the measure *Accuracy* is used to define correct recognition of each activity in a smart home. The measures *Precision* and *Recall* is used to understand *False\_Positive* and *False\_Negative*. When there is a poor balance between *Precision* and *Recall*, *F\_Measure* helps to understand the classifier performance.

#### 4.1 Results from Aruba17

This dataset was gathered from the test bed which is deployed in WSU, USA with the support of healthy and elderly woman. The result of the AR can be interpreted by making use of the confusion matrix illustrated in Table 2. The recognition measures of all the seven activities in Aruba17 dataset using SVM-R is illustrated in Table 3. Out of all activities "Bed\_to\_Toilet" activity is recognised accurately. Other two activities "Relax" and "Sleeping" are also recognised with commendable accuracy. At the average level of accuracy, the activities "Meal\_Preparation" and "Leave\_Home" are recognised. From the confusion matrix, one can observe that the activities "Eating", "Leave\_Home", "Meal\_Preparation", "Sleeping", and "Work" are wrongly recognised as "Relax". This can be understood by calculating Precision and Recall. Table 3 explains the classification performance of all seven activities on Aruba17 dataset using SVM-R classifier.

#### 4.2 Results from ShiB010

ShiB010 data set was collected from test bed home in India. This test bed is deployed using the sensors from Smart Home in a Box (ShiB) kit which was provided by CASAS project. Totally 59 volunteers aged between 33 and 83 years are participated in the study. At a time a single participant was involved in the data collection. The result of the AR can be interpreted with the support of confusion matrix shown in Table 2. The performance of the classifier SVM-R w.r.t recognising all seven activities on ShiB010 dataset is illustrated in Table 3. Out of all activities the four activities viz., "Bed\_to\_Toilet", "Eating", "Hygiene", and "Sleeping" activities are recognised accurately. Other three activities "Meal\_Preparation", "Relax" and "Work" are recognised with better level of accuracy. This can be understood by calculating Precision and Recall.

#### 4.3 Results from HH102

HH102 dataset was gathered from the test bed with the support of single resident. The result of the AR can be interpreted by using the confusion matrix shown in Table 2. The recognition measures of all the seven activities are

extracted in HH102 dataset using SVM-R. Out of all activities "Work" activity is recognised accurately in 85 percent of the cases.

Table 2. Confusion Matrix for recognition of all activities in Percentage for Datasets Aruba17, ShiB010, HH102, and HH113

| SVM-R (Aruba17)  |               |        |            |                  |       |          |      | SVM-R (ShiB010)  |               |        |         |                  |          |          |      |
|------------------|---------------|--------|------------|------------------|-------|----------|------|------------------|---------------|--------|---------|------------------|----------|----------|------|
|                  | Bed_to_Toilet | Eating | Leave_Home | Meal_Preparation | Relax | Sleeping | Work |                  | Bed_to_Toilet | Eating | Hygiene | Meal_Preparation | Relax    | Sleeping | Work |
| Bed_to_Toilet    | 100           | 0      | 0          | 0                | 0     | 0        | 0    | Bed_to_Toilet    | 100           | 0      | 0       | 0                | 0        | 0        | 0    |
| Eating           | 0             | 49     | 0          | 0                | 0     | 0        | 0    | Eating           | 0             | 100    | 0       | 0                | 0        | 0        | 0    |
| Leave_Home       | 0             | 0      | 67         | 0                | 0     | 1        | 0    | Hygiene          | 0             | 0      | 100     | 2                | 0        | 0        | 0    |
| Meal_Preparation | 0             | 28     | 2          | 76               | 1     | 2        | 5    | Meal_Preparation | 0             | 0      | 0       | 98               | 0        | 0        | 0    |
| Relax            | 0             | 24     | 31         | 24               | 98    | 10       | 55   | Relax            | 0             | 0      | 0       | 0                | 96       | 0        | 5    |
| Sleeping         | 0             | 0      | 0          | 0                | 0     | 87       | 0    | Sleeping         | 0             | 0      | 0       | 0                | 0        | 100      | 0    |
| Work             | 0             | 0      | 0          | 0                | 0     | 0        | 41   | Work             | 0             | 0      | 0       | 0                | 4        | 0        | 95   |
| SVM-R (HH102)    |               |        |            |                  |       |          |      | SVM-R (HH113)    |               |        |         |                  |          |          |      |
|                  | Bed_to_Toilet | Eating | Hygiene    | Meal_Preparation | Relax | Sleeping | Work |                  | Bed_to_Toilet | Eating | Hygiene | Meal_Preparation | Sleeping | Work     |      |
| Bed_to_Toilet    | 68            | 2      | 10         | 8                | 7     | 11       | 7    | Bed_to_Toilet    | 98            | 0      | 0       | 0                | 0        | 0        |      |
| Eating           | 0             | 60     | 0          | 0                | 0     | 0        | 0    | Eating           | 0             | 98     | 0       | 0                | 0        | 0        |      |
| Hygiene          | 13            | 0      | 73         | 5                | 4     | 14       | 5    | Hygiene          | 0             | 0      | 100     | 0                | 0        | 0        |      |
| Meal_Preparation | 1             | 18     | 1          | 57               | 5     | 3        | 2    | Meal_Preparation | 0             | 2      | 0       | 100              | 0        | 0        |      |
| Relax            | 1             | 9      | 1          | 2                | 52    | 1        | 1    | Sleeping         | 0             | 0      | 0       | 0                | 100      | 0        |      |
| Sleeping         | 0             | 0      | 0          | 0                | 0     | 57       | 0    | Work             | 2             | 0      | 0       | 0                | 0        | 100      |      |
| Work             | 16            | 11     | 16         | 28               | 31    | 13       | 85   |                  |               |        |         |                  |          |          |      |

Table 3. Performance Result of Individual Activity Classification of Aruba-17 and ShiB010 Dataset.

| Activities       | Aruba-17 |      |      |      |      | ShiB010 |      |      |      |      |
|------------------|----------|------|------|------|------|---------|------|------|------|------|
|                  | TP       | FP   | Pr   | Re   | F M  | TP      | FP   | Pr   | Re   | F M  |
| Bed_to_Toilet    | 1        | 0    | 1    | 1    | 1    | 1       | 0    | 1    | 1    | 1    |
| Eating           | 0.49     | 0.02 | 0.95 | 0.49 | 0.64 | 1       | 0    | 1    | 1    | 1    |
| Leave_Home       | 0.67     | 0.04 | 0.94 | 0.67 | 0.78 | -       | -    | -    | -    | -    |
| Hygiene          | -        | -    | -    | -    | -    | 1       | 0.02 | 0.98 | 1    | 0.99 |
| Meal_Preparation | 0.76     | 0.08 | 0.91 | 0.76 | 0.83 | 0.98    | 0    | 1    | 0.98 | 0.99 |
| Relax            | 0.98     | 0.1  | 0.91 | 0.98 | 0.94 | 0.96    | 0.04 | 0.96 | 0.96 | 0.96 |
| Sleeping         | 0.87     | 0.03 | 0.97 | 0.87 | 0.92 | 1       | 0    | 1    | 1    | 1    |
| Work             | 0.41     | 0    | 1    | 0.41 | 0.58 | 0.95    | 0.05 | 0.95 | 0.95 | 0.95 |

The accuracy level of other three activities "Bed\_to\_Toilet", "Eating" and "Hygiene" are 68%, 60%, and 73% respectively. However, the activities "Meal\_Preparation", "Relax", and "Sleeping" are recognised with less accuracy and many activities are recognised wrongly because of domination of activity "Work". This can be understood by calculating Precision and Recall. The performance of the classifier SVM-R w.r.t recognising all seven activities on HH102 dataset is illustrated in Table 4. Though, the accuracy of SVM-R w.r.t recognition of activities happened in HH102 is less, the F-Measure suggest that the performance is not too worse.

Table 4. Performance Result of Individual Activity Classification of HH102 and HH113 Dataset.

| Activities       | HH102 |      |      |      |      | HH113 |      |      |      |      |
|------------------|-------|------|------|------|------|-------|------|------|------|------|
|                  | TP    | FP   | Pr   | Re   | F M  | TP    | FP   | Pr   | Re   | F M  |
| Bed_to_Toilet    | 0.68  | 0.29 | 0.7  | 0.68 | 0.69 | 0.98  | 0    | 1    | 0.98 | 0.99 |
| Eating           | 0.6   | 0.01 | 0.92 | 0.6  | 0.73 | 0.98  | 0    | 1    | 0.98 | 0.99 |
| Leave_Home       | -     | -    | -    | -    | -    | -     | -    | -    | -    | -    |
| Hygiene          | 0.73  | 0.26 | 0.77 | 0.73 | 0.75 | 1     | 0    | 1    | 1    | 1    |
| Meal_Preparation | 0.57  | 0.1  | 0.77 | 0.57 | 0.66 | 1     | 0.01 | 0.99 | 1    | 1    |
| Relax            | 0.52  | 0.05 | 0.85 | 0.52 | 0.64 | -     | -    | -    | -    | -    |
| Sleeping         | 0.57  | 0.01 | 0.89 | 0.57 | 0.69 | 1     | 0    | 1    | 1    | 1    |
| Work             | 0.85  | 0.72 | 0.56 | 0.85 | 0.68 | 1     | 0    | 1    | 1    | 1    |

#### 4.4 Results from HH113

HH113 dataset was gathered from the test bed with the support of a single occupant. The result of the AR can be interpreted with the support of confusion matrix represented in Table 2. The recognition measures of all the six activities are calculated in HH113 dataset using SVM-R. Out of all activities "Hygiene", "Meal\_Preparation", "Sleeping", and "Work" activities are recognised accurately. The recognition accuracy level of other two activities "Bed\_to\_Toilet" and "Eating" are 98%. This can be understood by calculating Precision and Recall. The performance of the classifier SVM-R w.r.t recognising all six activities on HH113 dataset is illustrated in Table 4.

The result of four machine learning classifiers SVM-R, SVM-P, NB and KNN models are obtained using R-Programming, an open source tool. The evaluation metrics Accuracy, Precision, Recall and F\_Measure are used to measure the accuracy of activity recognition. The comparative performance of Area-Based segmentation on all four datasets is shown in Fig 4. In case of SVM, the selection of kernel type was crucial. Comparatively, RBF kernel classified better than polynomial kernel. In all four datasets, the SVM-R recognised all the activities and produced better result.

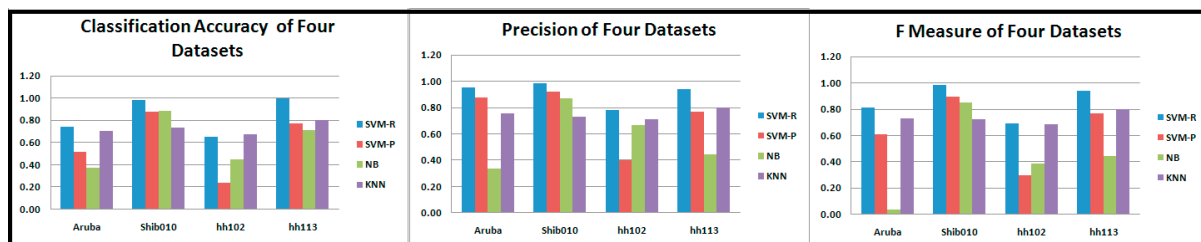


Fig 4. Comparative Performance of Area-Based Segmentation on Four Datasets.

## 5. Conclusion

This paper reviews the taxonomy of smart home activity recognition on ambient sensors. Activity recognition supports to monitor the resident who lives alone in a smart home. Segmentation plays a vital role in increasing the accuracy of recognising an activity. Area-Based segmentation divides the sensor data in to different segments based on the area inside the home and the deployment of the sensors in the smart home area. In this paper, different performance measures are used to conduct a comprehensive evaluation of Area-Based segmentation with the support of four different test bed data sets. Four different classifiers are used to evaluate the performance of Area-Based segmentation and test results prove that SVM-R recognises the activity with the highest level of accuracy as compared to the other three classifiers. As a future work, we will compare the performance of Area-Based segmentation with other segmentation techniques. We will also implement and test the area based segmentation in real time online smart home AR system to support disabled or aged smart home resident.

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